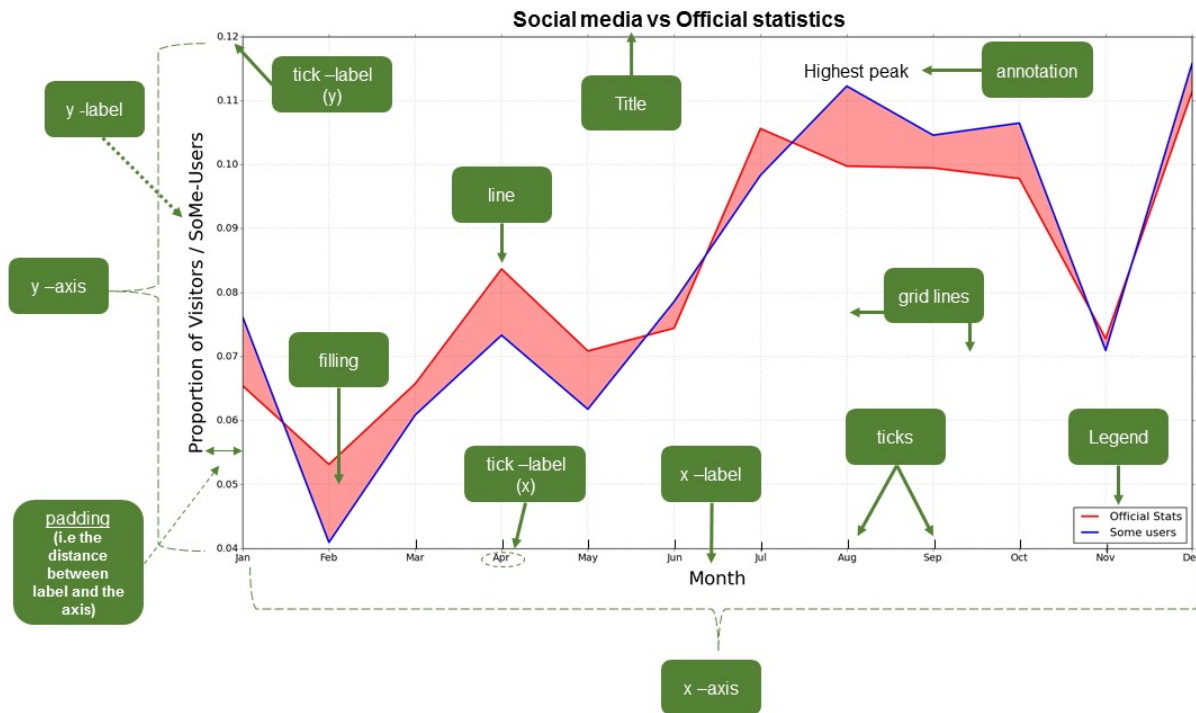


Anatomy of a Plot



Anatomy of a Plot

- axis Axis of the graph that are typically x, y and z (for 3D plots).
- title Title of the whole plot.
- label Name for to the whole axis.
- legend Legend for the plot.
- tick label Refers to the text or values that are represented on the axis.
- symbol Symbol for data point(s) (on a scatter plot) that can be presented with different symbols.
- size Size of e.g. a point on a scatter plot, also used for referring to the text sizes on a plot.
- linestyle The style how the line should be drawn. Can be e.g. solid or dashed.
- linewidth The width of a line in a plot.
- alpha Transparency level of a filled element in a plot (values between 0.0 (fully transparent) to 1.0 (no transparency)).
- tick(s) Refers to the tick marks on a plot.
- annotation Refers to the added text on a plot.
- padding The distance between a (axis/tick) label and the axis.

Important pyplot Methods

Task	pyplot method (plt....())
Generate & show a figure	figure(), show()
Making plots: lines bars, scatters	plot(), bar(), scatter()
Styling the graph	adjust plot() parameter
Adjusting axes	xlim(), ylim(), cla()
Labeling the plot or elements	title(), xlabel(), ylabel(), text()
Legends	legend() + add legend text as 'label' with plotted element
Making subplots	subplot(NMn)
Save image in various formats	savefig() (with figure instance)

Find more in the online documentation with pyplot:

https://matplotlib.org/api/pyplot_summary.html

<https://matplotlib.org/tutorials/introductory/pyplot.html>

Line Style, Marker, ...

linestyle	description
'-' or 'solid'	solid line
'--' or 'dashed'	dashed line
'-.' or 'dashdot'	dash-dotted line
':' or 'dotted'	dotted line

marker	description
''	point
','	pixel
'o'	circle
...	...

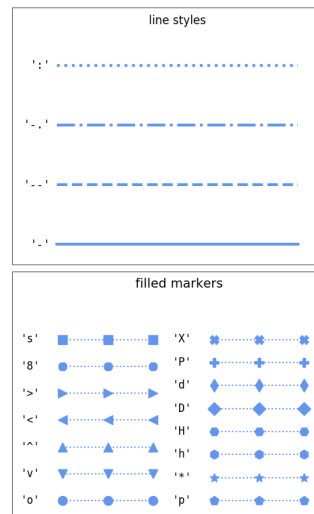
Also accepts drawstyles:

['default' | 'steps' | 'steps-pre' | 'steps-mid' | 'steps-post']

You can also:

- define own linestyles using set_dashes([])
- set style cycle withycler:

matplotlib.org/tutorials/intermediate/color_cycle.html#sphx-glr-tutorials-intermediate-color-cycle-py



Colors

Pyplot recognizes:

- RGB ('0.1, 0.2, 0.5')
- hex RGB ('#0F0F0F')
- one of {'b', 'g', 'r', 'c', 'm', 'y', 'k', 'w'};
- color sets names
 - x11/CSS4 ('blue')
 - xkcd ('xkcd:blue')



See more details:

<https://matplotlib.org/tutorials/colors/colors.html#sphx-glr-tutorials-colors-colors-py>

Save your Figure

```
# how to save your figure
import matplotlib.pyplot as plt
fig = plt.figure()
plt.plot(x,y)
fig.savefig('myplot.png')
plt.show() # optional & at last!!!
```

Supported formats:

```
{'eps': 'Encapsulated Postscript',
'jpeg': 'Joint Photographic Experts Group',
'jpg': 'Joint Photographic Experts Group',
'pdf': 'Portable Document Format',
'pgf': 'PGF code for LaTeX',
'png': 'Portable Network Graphics',
'ps': 'Postscript',
'raw': 'Raw RGBA bitmap',
'rgba': 'Raw RGBA bitmap',
'svg': 'Scalable Vector Graphics',
'svgz': 'Scalable Vector Graphics',
'tif': 'Tagged Image File Format',
'tiff': 'Tagged Image File Format'}
```